Technical approach

1 / Architecture and scope

Decision boundary

The prototype supports the Q4 2024 board cycle using public SEC filings dated on or before February 28, 2025, Dana's agreed preparation-date proxy. Reporting period and publication date remain separate. The supplied Apex file is disputed and withheld until a named approver and approval note are recorded; even after approval, rows after Q4 2024 cannot enter this historical comparison.

Data flow

Local workspace or Python CLI -> SEC submissions inventory, including historical pages -> eligible financial, portfolio, prospectus and current reports -> complete submissions and exhibits -> source cache -> metric-independent passage index -> question-specific evidence -> affected-cell LLM judgment -> deterministic citation/date/calculation validation -> HTML, Word, CSV, PDF and JSON. SEC sources use declared identification, a four-request-per-second ceiling, bounded retries and immutable SHA-256 content. Metric edits reuse indexed passages and preserve unrelated decisions; adding a fund processes only that fund. Three bounded workers review independent cells and checkpoint completed decisions. Reports read saved decisions and never invoke AI on download.

Extraction boundaries

BeautifulSoup/lxml parse inline XBRL, HTML tables and narrative. PDF text is extracted with pypdf. Selected NAV/book-value tags can qualify when units, instant date and undimensioned common-share context are explicit. Date-aligned financial highlights and total-return indices qualify only with matching class/basis and reinvestment evidence. Consecutive annual returns are compounded; rounded index levels support explicitly approximate calculations. Narrative rules collect candidate rates and entire supporting passages. Complex tables, class dimensions and ambiguous narratives require recorded review. N-PORT maps totals, borrowings, monthly returns and holdings with XML paths; assets minus liabilities is checked against net assets. Net assets are not NAV/share. Sparse PDF pages use local OCR with word boxes and mandatory visual review.

Financial definitions

For Golub and KREF, the primary return is shareholder total return; interval funds use net NAV returns for the intended class. These bases remain visibly different. Contractual management fees are primary; disclosed effective fees remain separate, with waiver terms in evidence. KREF is a mortgage REIT: book value is not labeled fund NAV, and leverage/yield require their stated definitions. No undocumented gross-to-net, leverage or yield conversions are performed. Supported return chains and index ratios retain inputs and formulas; quarterly fee/hurdle annualization preserves contractual conditions.

Implementation choice

Python handles retrieval and numeric validation; an OpenAI model interprets disclosures, selects or withholds values, and explains competing figures. Only public SEC passages enter model requests; disputed Apex data is excluded. Parser-only operation is explicitly labeled. Model credentials stay in ignored local configuration. The workspace searches SEC registrants and adds funds after the user specifies their structure and share class. PMs define custom metrics with units, scope and literal search phrases. These definitions bind model requests and caches. Word is the requested committee format; its comparison and evidence notes travel with the quarterly package.

Technical approach

2 / Evidence, judgment and reconciliation

Per-value evidence contract

Every candidate records ticker, metric, value if parsed, units, period, share class, basis, accession, filing date, original document URL, document hash, exact locator, source quote and extraction method. HTML uses source anchors, block/table ordinals or XBRL contexts instead of fabricated PDF page numbers. PDF passages use actual page numbers. Complete submissions and extracted exhibits are cached so results can be reproduced even if web presentation changes.

Evidence support and rationale

There are no ordinal confidence grades. Each result states whether evidence is linked, whether it is directly reported or calculated, why it was selected, and what remains unresolved. The LLM selects numbered passages and explains material competing figures. The application restores exact quotations and verifies passage IDs, source hashes, dates, units and implemented calculations locally. Linked evidence proves traceability, not semantic correctness or PM/compliance approval. A persuasive model statement cannot replace a source.

Conflicts and freshness

The model receives latest annual/interim and subsequent metric evidence; latest available and baseline sources are retained. Long blocks use excerpts with source offsets. Different figures require explicit explanations; unresolved contradictions remain blank. Truncated packets cannot support publication. Golub's 15.8% summary reconciles to its 15.77% detailed one-year shareholder return; the detailed figure is selected and both sources remain visible. Unresolved candidates block publication when their confirmed period and definition match and their value conflicts. Unscoped numeric passages instead flag the supported value for review, with the selection rationale visible. Explicit rejection requires reviewer identity and reason. Any retrieval failure marks fund coverage incomplete, preventing a stale value from appearing current. Period lag over 120 days is flagged as a configurable review threshold, not a business materiality policy.

Review controls

Decisions bind to source-derived candidate IDs. Acceptance requires a quoted substring, the numeric value present in that excerpt, compatible units, date, basis and class. Unsupported derived values cannot be manually introduced as extracted facts. OCR acceptance also requires recorded visual verification against the original page. Review is a local auditable file, not authenticated approval infrastructure. The output remains a draft until the PM and compliance independently approve it.

Validation performed and not performed

Tests use synthetic unit fixtures exclusively to exercise cutoff inclusion/exclusion, historical discovery, amendments/conflicts, units/scales, class ambiguity, evidence-bound decisions, internal-data withholding, and failure exports. Live SEC retrieval now works with a real declared contact. Live testing corrected N-PORT URL handling and stopped financial XML from being misclassified as prose. Checks cover Golub NAV, KREF book-value columns, current versus historical fee rates, quarterly annualization, return compounding, and the 15.8%/15.77% rounding reconciliation. This is not completed financial reconciliation: audit other actual passages, reconcile balances where definitions permit, inspect fee waivers and compare matching return periods before accepting broad coverage.

Technical approach

3 / Limitations, risks and rollout

Live-access validation

Initial requests returned HTTP 403. Providing a real declared contact enabled SEC retrieval. The application records HTTP status separately from filing identifiers, avoiding accidental classification of a URL as an access denial. N-PORT raw documents are fetched from the accession root rather than the SEC display-folder path. Parsed evidence is cached by source, configuration and parser version. Retrieval success is not a financial correctness certificate.

Extraction and normalization risk

Custom metrics currently accept reported values only, with no arbitrary formulas or custom XML mapping. SEC ticker directories are incomplete; name/CIK fallback and series/class review are needed. The generic rules cover fewer layouts than a production service. Narrative percentages can refer to expense ratios, old terms or thresholds; they remain candidates. Institutional class mappings need confirmation from actual filings. OCR handles sparse-text scans; mixed pages can force OCR. Missing engines, timeouts and unreadable pages become explicit errors. OCR still requires visual review. N-PORT borrowing scope and unidentified monthly return classes remain review items; derivative contracts and risk grids are not normalized. Model judgments can misinterpret financial language; validate actual responses and audit representative cases. There is no external market-price feed. Return calculations use only disclosed annual returns or reinvested-dividend indices. Golub's same-period NAV/shareholder return difference is shown. Cliffwater's partial fiscal transition prevents automatic return chaining; incomplete history and unresolved definitions remain gaps.

Source-selection risk

Broad historical report and recent supplement discovery reduces missed exhibits but can be expensive. Current metadata filtered by filing date is not a separately archived 2025 snapshot. Filing and amendment availability should be checked against the SEC index during calibration. Conservative fund-wide blocking after a failed document can reduce usable coverage; production should track which metrics depend on each missing source without silently weakening freshness claims.

Production safeguards

Add per-fund regression documents, extraction coverage measures, authenticated reviewer roles, immutable decision versions and a PM/compliance release gate. Keep a filing inventory diff on each run to detect new amendments. Store raw sources in versioned object storage, use a shared rate limiter and alert on SEC denials, source gaps and definition changes. Client inputs require access controls and approved provenance. Keep outputs private until authorized.

Delivery and next checkpoint

The repository supplies installation, CLI, local evidence review, portable exports, tests, decisions and the business brief. Use a representative sample of the downloaded results to quantify correct extraction and review burden. Bring a real fee conflict and the KREF comparison to Dana in the next consultation. Validate the requested Word package and resolve internal data before staging a PM review. Scheduling and hands-off operation should follow successful quarterly parallel runs, with acceptance criteria agreed by the client.